Vacuum Polarization in 1+1 dimensions

Authors

Abstract

In this article...

Contents

1	Introduction	1
2	Set up	2
2.1	Field modes	2
2.2	External Field Approximation and Critical Field Strength	4
2.3	Mode sum formula	4
3	Self-consistent fields	5
4	Results	6
4.1	Mode sum formula	6
5	Discussion and outlook	8
A	Computing the vacuum polarization	8

1 Introduction

Vacuum polarization is a phenomenon in quantum electrodynamics (QED) by which the vacuum of a charged field polarizes when subject to external electric fields, effectively behaving like a dielectric with $\epsilon > 0$. This mechanism was one of the first QED effects studied [3], with consequences that followed closely [14; 6]. This mechanism has been already observed, e.g. in corrections to atomic levels [9] (cf. Ref. [10] for a review), while other effects e.g. the Schwinger effect [12] lay beyond current technological advances (cf. Ref. [4]).

Defining vacuum polarization for generic backgrounds[1] is no easy task. Historically, [3] first defined it through the assumption that the short-distance behavior of the physically relevant states in the presence of strong background electric fields is the same as in their absence, only up to smooth coefficients. In modern bibliography, these states are called Hadamard, (cf Ref. [5]), and they are particularly interesting in the context of quantum field theory on curved spacetimes, where the same idea is used to define observables necessary for the study of e.g. perturbative interacting field theory or semiclassical gravity (cf. [7; 8] for detailed discussions on the definition on such observables).

Semiclassical gravity, a possible first step towards a quantum theory of gravity, couples classical gravity to the expectation value of the stress-energy tensor, which similarly as the vacuum polarization needs to be redefined. To our knowledge, only perturbative corrections to the metric have been studied [13]. In this article we study, as a toy model, the effect of the backreaction of a charged scalar field to a background classical electric field in 1+1 dimensional Minkowski spacetime. This setup was already studied in [1], where it was found that ignoring the backreaction of the scalar field leads to instabilities, and it claims that the inclusion thereof avoids them. However, as pointed out in [11], the expression used in [1] for vacuum polarization ρ cannot be derived from a manifestly gauge invariant prescription. [15] derives the correct expression for ρ, which has qualitative disagreements with the one used in [1].

This discussion motivates the following question, which we answer in the present paper: Given the difference in the prescriptions for ρ, does using the correct one avoid these instabilities? We will use the correct prescription described in [15] to implement an iterative method to study the effect of the backreaction of the scalar field. It has been found that this expression also alleviates the above mentioned instabilities.

The layout of this article is as follows: Section 2 ...

Maybe add?: [2] discusses hadamard states in the pressence of time-like boundaries.

[1] And any other observable defined through the evaluation of products of fields or their derivatives at coinciding points.

This article summarizes and extends the results of the M.Sc. thesis of Santiago Sanz-Wuhl, written in 2025 at the Institut für Theoretische Physik, Universität Leipzig, under the supervision of Jochen Zahn.

2 Set up

In 1+1 dimensional spacetime, two charges $q, -q$ are set at $x^1 = 0, x^1 = a$, respectively. Ignoring vacuum polarization effects, this yields a constant electric field of strength q pointing towards positive x^1 in the region $x^1 \in [0, a]$, 0 everywhere else. We study the solutions to the semiclassical Klein-Gordon-Maxwell equations (KGM), i.e. the system of coupled differential equations

$$\left(D_\mu D^\mu + m^2\right)\phi = 0, \tag{1a}$$

$$\partial_\mu F^{\mu\nu} = \langle j^\nu \rangle, \tag{1b}$$

in the region $x = (x^0, x^1) \in \mathbb{R} \times [0, a]$. Here, $D_\mu = \partial_\mu + ieA_\mu$ is the gauge covariant derivative,

$$j^\nu(x) = \frac{ie}{2}\left((D^\nu\phi^*(x))\phi(x) - \phi^*(x)D^\nu\phi(x)\right) \tag{2}$$

is the charge current density of the Klein-Gordon field, and $\langle \mathcal{O} \rangle$ is the vacuum expectation value of an observable $\mathcal{O}$, and the metric signature is $(+, -)$. We solve for $F_{\mu\nu}$ and (the mode decomposition of) the two-point function of the vacuum state.

In 1+1 dimensions, A_1 can be arbitrarily set to 0. This way, Eq. (1b) reduces to the Poisson equation for the electrostatic potential A_0

$$\partial_1^2 A_0(x^1) = -\left\langle j^0 \right\rangle \equiv -\rho, \tag{3}$$

with ρ the vacuum polarization, i.e. the vacuum expectation value of the zeroth component of the charge current density. Eq. (3) will be solved with the boundary conditions

$$\partial_1 A_0(x^1)\big|_{x^1=0,a} = -q. \tag{4}$$

This way, measuring the electric field close enough to the boundary is not affected by vacuum polarization. The boundary conditions (4) fix the solution to Eq. (3) up to an addition constant, which is for convenience chosen so that $A_0(\frac{a}{2}) = 0$. With these boundary conditions and for a given $\rho(x^1)$, Eq. (3) is solved by

$$A_0(x^1) = -q\left(x^1 - \frac{a}{2}\right) - \int_{\frac{a}{2}}^{x^1}\int_0^\chi \rho(\xi)d\xi d\chi. \tag{5}$$

2.1 Field modes

In this gauge, the Klein-Gordon equation is of the form

$$\left[(\partial_0 + ieA_0)^2 + \partial_1^2 - m^2\right]\phi = 0, \tag{6}$$

which we will study using Dirichlet boundary conditions (DBC) $\phi|_{x^1=0,a} = 0$ or Neumman boundary conditions (NBC) $\partial_1\phi|_{x^1=0,a} = 0$. More general boundary conditions may be considered, though they are outside of the scope of this work. Using the mode ansatz $\phi_n(x) = \phi_n(x^1)e^{-i\Omega_n x^0}$ the PDE is split into ordinary differential equations for each mode $\phi_n(x^1)$ of the scalar field

$$\left[(\Omega_n - eA_0)^2 + \partial_1^2 - m^2\right]\phi_n = 0. \tag{7}$$

The following dimensionless parameters are defined

$$z = a^{-1}x^1 \tag{8a}$$

$$\lambda = eEa^2 \tag{8b}$$

$$\varepsilon = ae \tag{8c}$$

$$\omega_n = a\Omega_{n,\alpha} \tag{8d}$$

Eq. (7) then takes the form

$$\left[(\omega_n - \varepsilon A_0)^2 + \frac{d^2}{dz^2} - a^2m^2\right]\phi_n = 0. \tag{9}$$

The modes ϕ_n are normalized with respect to the symplectic norm

$$(\phi_n, \phi_m) = i\int_0^1 (\phi_n^*\pi_m^* - \pi_n\phi_m)dz, \tag{10}$$

with $\pi_n = (D_0\phi_n)^*$ the canonical conjugate of ϕ_n.

The general solution to Eq. (6) is thus decomposed in modes

$$\phi(t, z) = \sum_{n>0} a_n\phi_n(z)e^{-i\omega_n t} + \sum_{n<0} b_n^\dagger\phi_n(z)e^{-i\omega_n t}. \tag{11}$$

Positive (negative) values of n are identified with modes of positive (negative) norm. a_n and b_n are the ladder operators for the positive and negative norm solutions, respectively, obeying canonical commutation relations $[a_n, a_n^\dagger] = [b_n, b_n^\dagger] = \delta_{nm}$, and all other commutators vanishing. The ladder operators define the vacuum state $|0\rangle$ as that for which $a\,|0\rangle = b\,|0\rangle = 0$. Note that due to the chosen gauge, $\omega_n = -\omega_{-n}$, and the $b_n^\dagger$ are the creation operators of the negative frequency family of solutions.

As mentioned above, the definition of quantities such as ρ —Wick polynomials—is problematic, since it involves the product of local observables evaluated at coinciding spacetime points (see Eq. (2)). To define such polynomials, one assumes that the states of a field ϕ are Hadamard, i.e. they have a similar short-distance behavior to the vacuum of ϕ in absence of background fields, and are assumed to be the only physically relevant states [5]. Under this assumption, the divergence of such states is state independenent and locally constructed from the background fields. Since ϕ is coupled to the background field through the covariant derivative, one may use the setup at hand as a toy model for semiclassical gravity, a possible first step towards an elusive quantum theory of gravity.

Specifically, one says that the two point function $w_\Omega^{\phi\phi^*}(x,x') = \langle\Omega|\phi(x)\phi^*(x')|\Omega\rangle$ of the scalar field ϕ in a Hadamard state Ω is of the form

$$w_\Omega^{\phi\phi^*}(x,x') \equiv H^{\phi\phi^*}(x,x') + R_\Omega^{\phi\phi^*}(x,x'). \tag{12}$$

Here,

$$H^{\phi\phi^*} = -\frac{V(x,x')}{4\pi}\ln[(t-t')^2-(z-z')^2+i\varepsilon\mathrm{sign}(t-t')] \tag{13}$$

is the Hadamard parametrix of the scalar field in 1+1 spacetime dimensions and $R_\Omega^{\phi\phi^*}$ a state-dependent smooth function. The smooth coefficient V, up to order relevant for vacuum polarization reads [11]

$$V(x,x') = \exp\left[-i\varepsilon\int_0^1 A_\mu(x'+s(x-x'))(x-x')^\mu ds\right] + \mathcal{O}(x-x')^2.$$

The Hadamard condition allows us to define expectation values of local observables non-linear in the fields

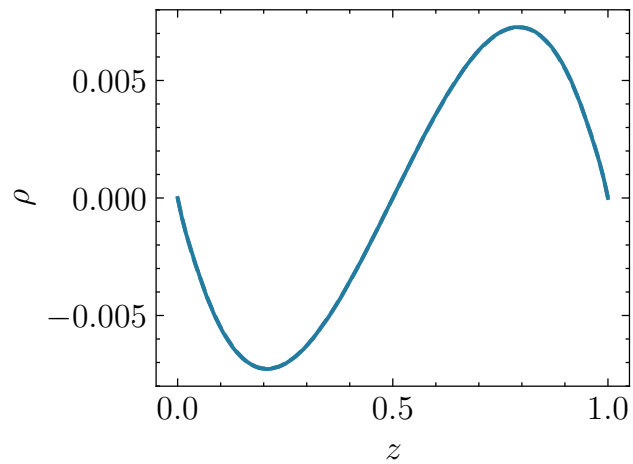

Fig. 1: The vacuum polarization ρ of the scalar field due to a background electrostatic potential $A_0(z) = -0.5(z-0.5)$, with DBC and $m = 0$.

as a limit of coinciding points:

$$\begin{aligned}&\langle D_\alpha\phi(x)D_\beta^*\phi^*(x)\rangle \\ &\quad = \lim_{x'\to x} D_\alpha D_\beta^{*\prime}(w_\Omega^{\phi\phi^*}(x,x') - H^{\phi\phi^*}(x,x')),\end{aligned} \tag{14}$$

with α, β symmetrized multiindices. The above prescription is called *point-splitting with respect to a Hadamard parametrix.*

Eqs. (14), (2) and the mode decomposition (11) together result in the expression for ρ calculated in [15]

$$\begin{aligned}\rho = \lim_{\tau\to 0}\sum_{n>0}\Big[&(\omega_n - \varepsilon A_0)|\phi_n|^2 e^{-i\omega_n\tau} \\ &+ \sum_{n<0}(\omega_n - \varepsilon A_0)|\phi_n|^2 e^{i\omega_n\tau}\Big] + \frac{\varepsilon^2}{\pi}A_0.\end{aligned} \tag{15}$$

An example of ρ using this expression is given in Fig. 1, calculated in the external field approximation explained below, i.e. a homogeneous background electric field given by the potential $A_0(z) = -0.5(z-0.5)$.

To measure the *strength* of the vacuum polarization, the *induced charge*

$$Q_I = \int_0^{\frac{1}{2}} \rho(z)dz \tag{16}$$

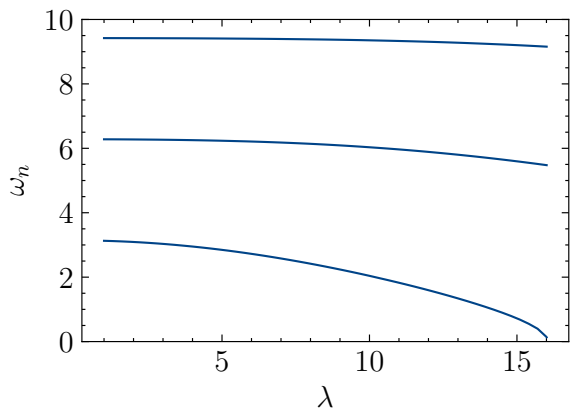

Fig. 2: Energies of the three first modes of the massless Klein-Gordon field subject to DBC in the external field approximation as functions of λ. $\omega_1 \to 0$ as $\lambda \to \lambda_c \approx 16$ defines the critical field strength for these mass and boundary conditions configurations.

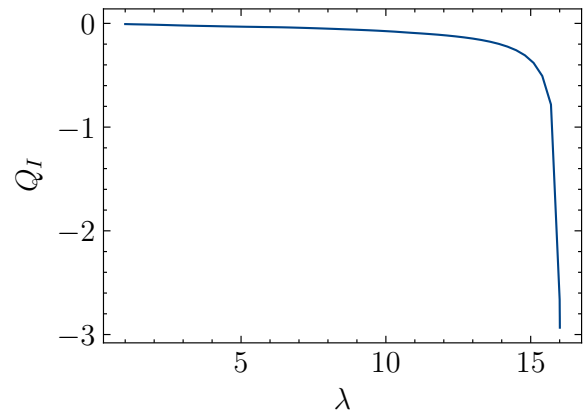

Fig. 3: Corresponding induced charge Q_I in Fig. 2 as a function of λ.

is defined. This charge partially screens the background electric field λ, and Gauß's law yields that the total electric field at $z = \frac{1}{2}$ as $\lambda + Q_I$.

2.2 External Field Approximation and Critical Field Strength

The external field approximation ignores the contribution of the backreaction of the charged scalar field to the background electric field. This approximation displays the instabilities of the system, which the inclusion of backreaction aims to avoid. Eq. (9) is solved with $A_0(z) = -\lambda(z - \frac{1}{2})$, and takes the form

$$\left[\left(\omega_n + \lambda\left(z - \frac{1}{2}\right)\right)^2 + \frac{d^2}{dz^2} - a^2 m^2\right]\phi_n = 0. \quad (17)$$

For each value of λ, the modes ϕ_n and their energies ω_n can be obtained from the boundary value problem. Fig. 2 displays the energies $\omega_1, \omega_2, \omega_3$ of the first three modes of the massless scalar field with DBC as functions of the background field strength λ. The value at which ω_1 goes to 0 defines the critical field strength $\lambda_c \approx 16.05039$. Fig. 3 displays the divergent behavior of $Q_I(\lambda)$ for this configuration as $\lambda \to \lambda_c$. λ_c adopts different values for different boundary conditions and masses.

2.3 Mode sum formula

Vacuum polarization in [1] is defined through a so-called mode sum formula. It pairs the charge density associated to each mode as

$$\rho_n = (\omega_n - \varepsilon A_0)|\phi_n|^2 - (\omega_n + \varepsilon A_0)|\phi_{-n}|^2. \quad (18)$$

to define the vacuum polarization

$$\begin{aligned}\rho^{\mathrm{MS}} &= \sum_{n>0} \rho_n \\ &= \sum_{n>0} \left[(\omega_n - \varepsilon A_0)|\phi_n|^2 - (\omega_n + \varepsilon A_0)|\phi_{-n}|^2\right],\end{aligned} \quad (19)$$

where the superscript MS is introduced to differentiate it from the correct prescription ρ. The above expression cannot be derived from a manifestly gauge invariant renormalization scheme, and it relies on the specific pairing of modes to be well-defined. Furthermore, in the iterative procedure discussed above, [1] truncates the mode sum to $\rho^{\mathrm{MS}} \approx \rho_1$. It is however worth pointing the reader that Eq. (15) is effectively a mode sum formula with the additional $\frac{\varepsilon^2}{\pi} A_0$ term, as discussed in Appendix A.

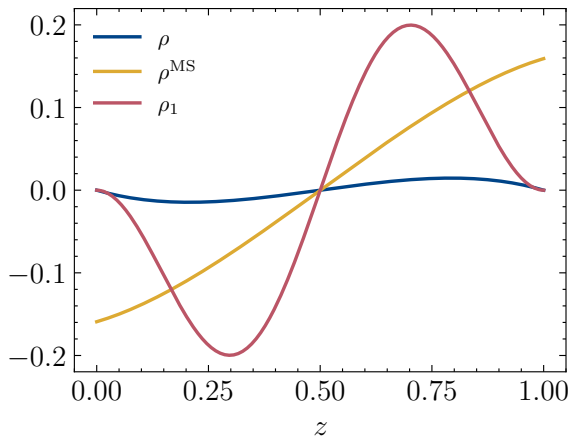

Fig. 4: Three different ρ for the same massless scalar field subject to DBC resulting from $A_0 = -(z-\frac{1}{2})$: In blue, the correct prescription in Eq. (15) used throughout this paper. In orange, ρ calculated through the mode sum (19). In red, ρ defined through the mode sum formula, truncated to the first mode.

Fig. 4 shows different charge densities arising from the different ρ prescriptions, for the massless scalar field subject to DBC. The same ϕ_n, ω_n and $A_0(z) = -(z-\frac{1}{2})$ have been used for the different ρ. In blue, the vacuum polarization used in the computations throughout this paper, in orange the vacuum polarization arising from the mode pairing and in red the one used for the iterative procedure in [1]. Notice the non-zero vacuum polarization at the boundaries for Eq. (19), which the reader might find surprising to find in a field subject to DBC. This is an artifact of the missing $\frac{\varepsilon^2}{\pi}A_0$ term in Eq. (19).

We observe that ρ is significantly 'weaker' than the one used by [1], where it was seen that the screening due to ρ^{MS} avoids the instabilities appearing in the external field approximation. This motivates the question answered in this paper: Does considering the correct vacuum polarization ρ also avoid these instabilities?

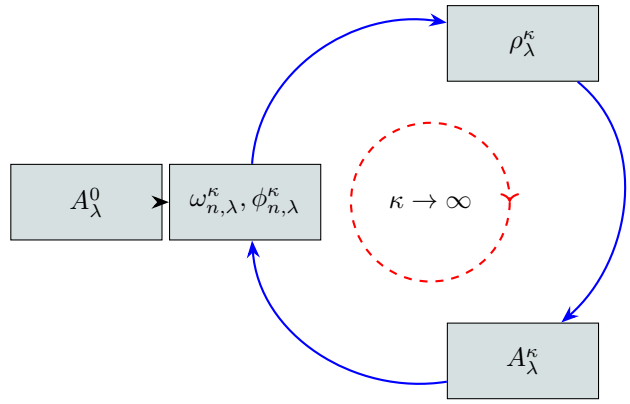

Fig. 5: Schematic of the iterative procedure for fixed λ.

3 Self-consistent fields

As discussed above, the polarization of the vacuum induces a charge density ρ, which modifies the background field as in Eq. (5). This modified background field can be used in Eq. (20a) to calculate new solutions, that will have a different vacuum polarization. This iterative procedure, schematized in Fig. 5, is performed *ad infinitum*, and we study its convergence.

More precisely, for each value of λ and at each iteration κ, we compute the following

$$\frac{d^2\phi^\kappa_{n,\lambda}}{dz^2} = -\left[(\omega^\kappa_{n,\lambda} - \varepsilon A^{\kappa-1}_{0,\lambda})^2 - a^2m^2\right]\phi^\kappa_{n,\lambda}, \tag{20a}$$

$$\begin{aligned}\rho^\kappa_\lambda = \lim_{\tau\to 0}\Big[&\sum_{n>0}(\omega^\kappa_{n,\lambda} - \varepsilon A^{\kappa-1}_{0,\lambda})|\phi^\kappa_{n,\lambda}|^2 e^{-i\omega^\kappa_{n,\lambda}\tau} \\ &+ \sum_{n<0}(\omega^\kappa_{n,\lambda} - \varepsilon A^{\kappa-1}_{0,\lambda})|\phi^\kappa_{n,\lambda}|^2 e^{i\omega^\kappa_{n,\lambda}\tau}\Big] \\ &+ \frac{\varepsilon^2}{\pi}A^{\kappa-1}_{0,\lambda},\end{aligned} \tag{20b}$$

$$\begin{aligned}A^\kappa_{0,\lambda} &= cA^{\kappa-1}_{0,\lambda} \\ &+ (1-c)\left[-\lambda\left(z-\frac{1}{2}\right) - \int_{\frac{1}{2}}^{z}\int_0^{\chi}\rho^\kappa_\lambda(\xi)d\xi d\chi\right].\end{aligned} \tag{20c}$$

This procedure has two main differences from the one used in [1]: the expression for the vacuum polar-

ization, and the $c < 1$ prescription in Eq. (20c), which is discussed below. The details on how ρ is calculated are explained in Appendix A. Initial conditions of this procedure are discussed below.

Noticing how Eq. (20c) acts as an update function on $A^{\kappa-1}_{0,\lambda}$, we write this scheme as an infinite dimensional fixed point problem. This is, for each λ, we look for the fixed point $A_{0,\lambda}$ obeying

$$A_{0,\lambda} = f_\lambda(A_{0,\lambda}), \tag{21}$$

with f_λ given by the composition of steps (20a), (20b) and (20c). The fixed points of f_λ, denominated "self-consistent solutions", are looked for by studying the limit $A_{0,\lambda} = \lim_{\kappa\to\infty} A^\kappa_{0,\lambda}$, where

$$A^{\kappa+1}_{0,\lambda} = f_\lambda(A^\kappa_{0,\lambda}) = f^\kappa_\lambda\left(A^0_{0,\lambda}\right), \text{with } f^\kappa_\lambda = \underbrace{f_\lambda \circ ... \circ f_\lambda}_{k \text{ times}}. \tag{22}$$

If the limit $A_{0,\lambda}$ exists, it is given the name self-consistent potential, as well as the corresponding self-consistent vacuum polarization ρ_λ, field modes $\phi_{n,\lambda}$ and mode energies $\omega_{n,\lambda}$.

For low enough $\lambda \ll \lambda_c$, the initial value of the sequence $A^0_{0,\lambda}$ can be set to the external field approximation $A^0_{0,\lambda}(z) = -\lambda\left(z - \frac{1}{2}\right)$. This is, however, no longer a valid initial potential for stronger background fields, as it leads to non-convergences in the sequence $A^\kappa_{0,\lambda}$. This is discussed in the Appendix ??(to be added). For higher λ-values, $A^0_{0,\lambda}$ is instead set by a stored previously calculated self-consistent potential $A_{0,\lambda'}$ with $\lambda = \lambda' + \Delta\lambda$, where $\Delta\lambda > 0$. We use this scheme to probe the λ-parameter space.

Finally, in Eq. (20c), $c \in [0, 1]$ is introduced in order to avoid an overshooting of the potential: For $c = 0$, the procedure is precisely the one used in [1], and it is expected to quickly converge. For c close to 1, updates will be small corrections to the previous potential. Stronger background field strengths increase the instability of the coupled system, thus increasing the relevance of slower converge schemes. Considering this, we adopt a scheme in which, as λ increases, c increases and simultaneously $\Delta\lambda$ decreases.

4 Results

Using the iterative procedure above laid out we focus on two particular cases: The massless scalar field subject to DBC, and $m = 1$ with the scalar field subject to NBC. We do not study the massless field subject to NBC, since under these boundary conditions, $\lambda_c = 0$ for $m = 0$, as pointed out in Ref. [15].

Figures 6 and 7 display the dynamics of KGM system in the semiclassical approximation. The external field approximation is given for reference in orange. Notice the different λ scales. The system is studied through the energies of the first modes (top panel), the induced charge Q_I (middle panel) and the total electric field $\lambda + Q_I$ at the midpoint $z = 0.5$ (bottom panel). We observe in both cases how considering backreaction avoids the instabilities reflected in $\omega_1 \to 0$ and $Q_I \to -\infty$ as $\lambda \to \lambda_c$ that the external field approximation presents. Indeed, the energy of the lower modes is raised which "prevents" ω_1 from crossing 0. We also point the reader to notice how the rapid growth of $|Q_I|$ as λ crosses λ_c keeps $\lambda + Q_I$ below λ_c. We find this behavior surprising, as intuitively one would expect $\lambda + Q_I$ to stabilize. Considering a quantum background electric field would perhaps fulfill this intuition.

One might also notice the almost identical behavior of the self-consistent solutions to the external field approximation in the weak-field regime $\lambda \ll \lambda_c$ in both cases.

4.1 Mode sum formula

For completeness, we compare the results obtained in this work with those obtained in [1]. Fig. 8 displays Q_I in the iterative procedure as a function of λ for two different prescriptions for ρ. In blue, the correct prescription ρ used throughout this paper, and in orange ρ^1 as discussed in 2.3, truncated to the first mode. Notice the different behaviors in the two regimes: although the truncated mode sum formula presents a noticeably stronger vacuum polarization in the weak-field regime, the strong-field regime behavior is very similar. This might be understood as a result of the lower modes being the most sensitive to background fields, and thus those who contribute the most to vacuum polarization.

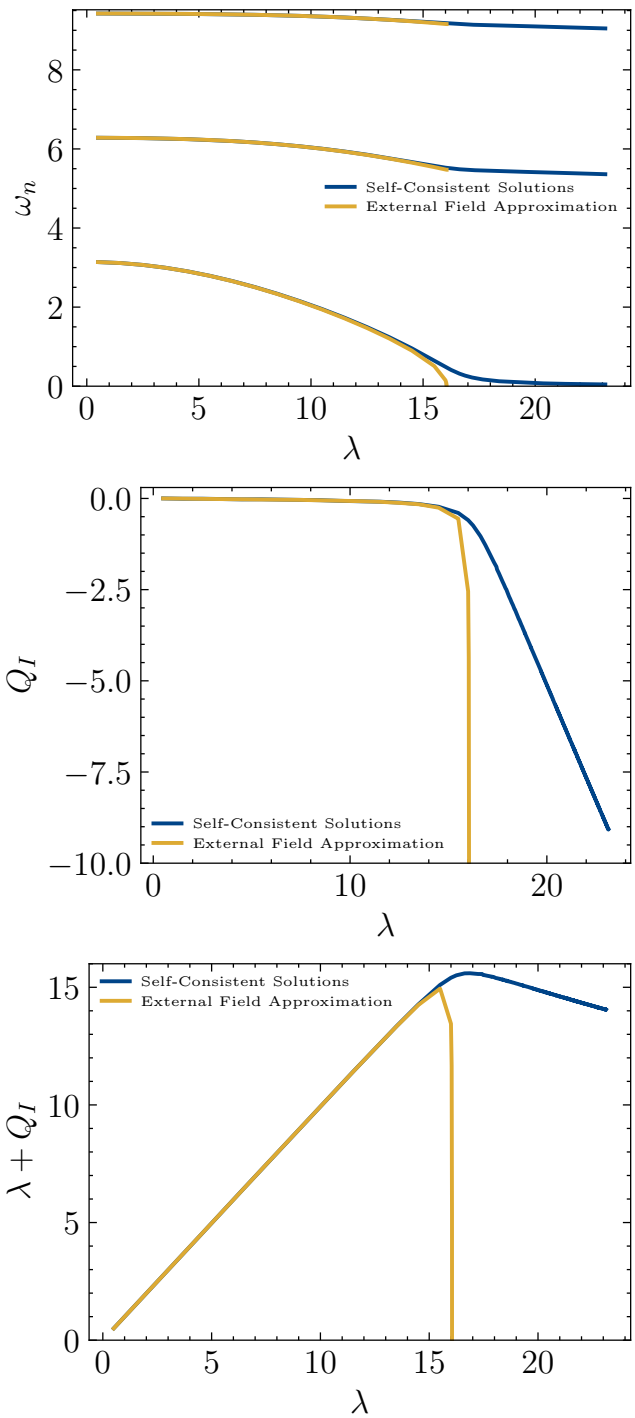

Fig. 6: In blue, the self-consistent solutions to the coupled semiclassical Klein-Gordon-Maxwell system for a massless scalar field subject to DBC. In orange, for reference, the external field approximation is given. Top panel: Energies of the first three modes as functions of λ. Middle panel: Corresponding Q_I for each λ. Bottom panel: Total electric field $\lambda + Q_I$ due to the screening of the vacuum at $z = \frac{1}{2}$.

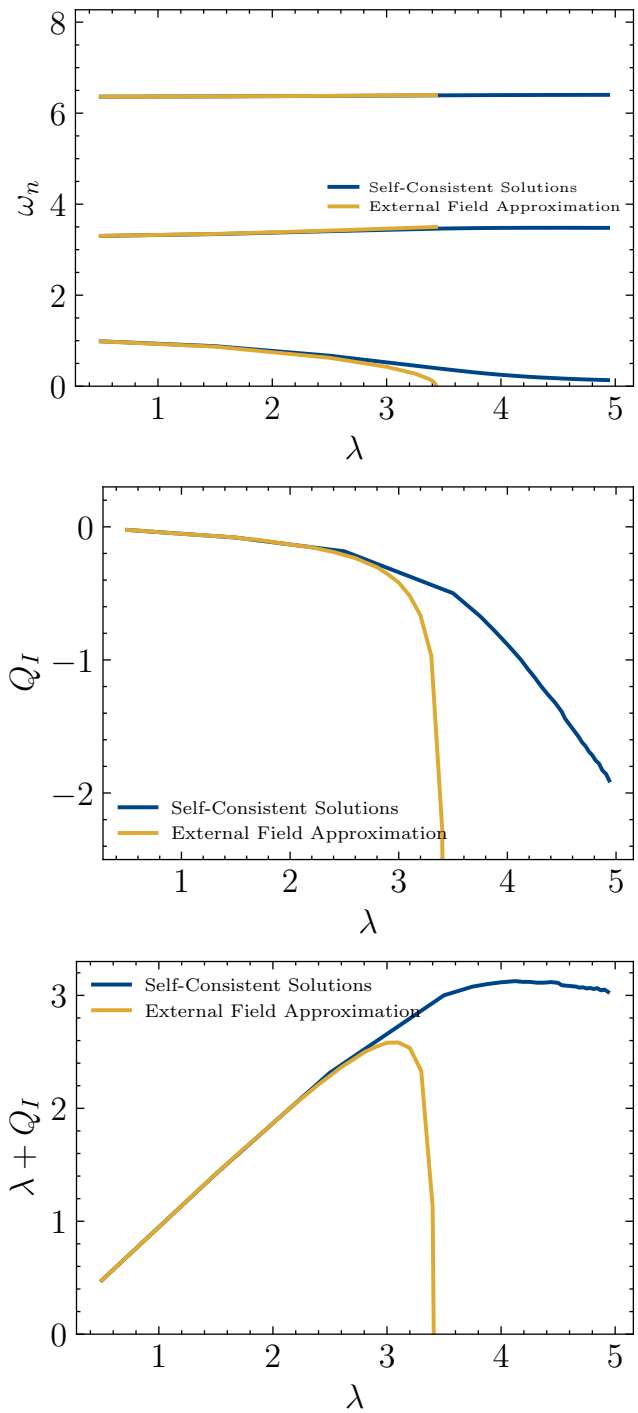

Fig. 7: Following the format in Fig. 6, the results for the unit mass scalar field subject to NBC. Further details in main text.

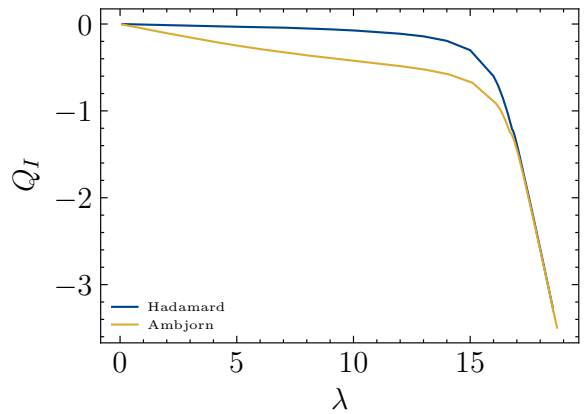

Fig. 8: Q_I for the self-consistent solutions for a massless field subject to DBC, by using different expressions for ρ. In blue, the (correct) vacuum polarization obtained through renormalization through point-splitting w.r.t. a Hadamard parametrix, and in orange the vacuum polarization obtained through a mode sum formula truncated to one mode used in [1].

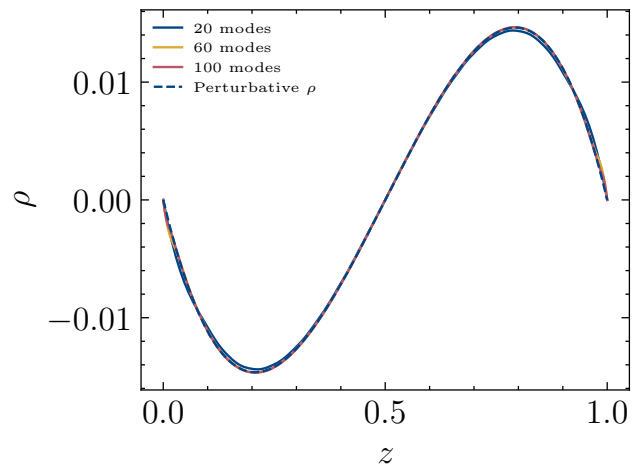

Fig. 9: ρ of the massless complex scalar field subject to a constant background electric field of strength $\lambda = 1$, for different mode cut-offs, as indicated in the legend. For reference, the perturbative vacuum polarization is given.

5 Discussion and outlook

We have studied the semiclassical Klein-Gordon-Maxwell system equations through an iterative procedure to study the effect of backreaction to a background classical electric field with a well-defined prescription of vacuum polarization. This precise set up was already studied in [1], where vacuum polarization is calculated using a mode sum formula, expression that cannot be derived from gauge invariant renormalization procedures.

A Computing the vacuum polarization

To be able to compute Eq. (15), some rearrangements are to be done. By use of $-\omega_n = \omega_{-n}$, ρ can be expressed as a single sum

$$\rho = \lim_{\tau\to 0} \sum_{n>0} \Big[(\omega_n - \varepsilon A_0)|\phi_n|^2 - (\omega_n + \varepsilon A_0)|\phi_{-n}|^2 \Big] e^{-i\omega_n \tau} + \frac{\varepsilon^2}{\pi} A_0(z) \tag{23}$$

In general, the limit and the sum do not commute. In order for them to commute, the coefficients of the complex exponentials need to decay fast enough, in order for the series to be absolutely convergent. After commuting, the sum is cut off at some finite mode N, which induces oscillations of frequency $\Delta z = \frac{1}{N+1}$. These oscillations are an artifact of the cut off, which must be smoothened out by a low-pass filter.

To achieve this, a low-pass filter is applied to the unfiltered vacuum polarization, by convoluting it with a normalized gaussian function, of approximately 3 oscillations of width. The ρ resulting from the filtering is displayed in Fig. 9, where the filtering has been applied to the vacuum polarization resulting from different mode cut-offs (20, 60 and 100, as indicated by the legend). For reference, the vacuum polarization calculated from perturbative theory as per [15] is also given.

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